STUDENTS INTERNSHIP REPORT
&
FEEDBACK FORM

DEPARTMENT OF INFORMATION TECHNOLOGY

Submitted by

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INDUSTRY NAME (address) : **EMGLITZ TECHNOLOGIES SAIBABA COLONY,COIMBATORE**

DATE : **30/12/2024 To 18/01/2025**

No. of days : **22**

HINDUSTHAN INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi, Accredited with 'A' Grade by NAAC Coimbatore 641032.

HINDUSTHAN INSTITUTE OF TECHNOLOGY

(*An Autonomous Institution)*

COIMBATORE

ACADEMIC YEAR 2024-2025

Department of Information Technology

NAME OF THE INDUSTRY & ADDRESS	:	**EMGLITZ TECHNOLOGIES PVT LTD SAIBABA COLONY,COIMBATORE**
DATE	:	**30/12/2024 to 18/01/2025**
ACADEMIC YEAR	:	**2024 - 2025**

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Coimbatore 641032.

BONAFIDE CERTIFICATE

This is to certify that the internship work submitted in partial fulfillment of the requirement for the award of the Degree of Information Technology is the Bonofide record of the internship work done by **KARNAN K- 720823106039** during the academic year 2024 - 2025.

INTERNSHIP COORDINATOR	**HEAD OF THE DEPARTMENT**
Mrs. V .Umashankari M.E	**Dr.M.Duraipandian M.E.,Ph.D.**
Assistant Professor	Professer and Head
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Submitted for the Practical Examination of the Anna University conducted on

…….

Internal Examiner External Examiner

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I am writing this final evaluation report as part of the Internship Program of the Functional Departments. This report includes **Emglitz Technologies, Coimbatore.**

It has been a great challenge but plenty of learning and opportunities to gain huge knowledge on the way writing this report. I express my sincere thanks to **Dr.M.Duraipandian M.E.,Ph.D.,** our amazing HOD of Information Technology department, who seemed to be with us always; and prepared to give us feedback and guidelines whenever we needed it.

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JAVA

INTERNSHIP

REPORT

INTERNSHIP REPORT ON JAVA

A brief explanation of JAVA programming language and its real-time applications

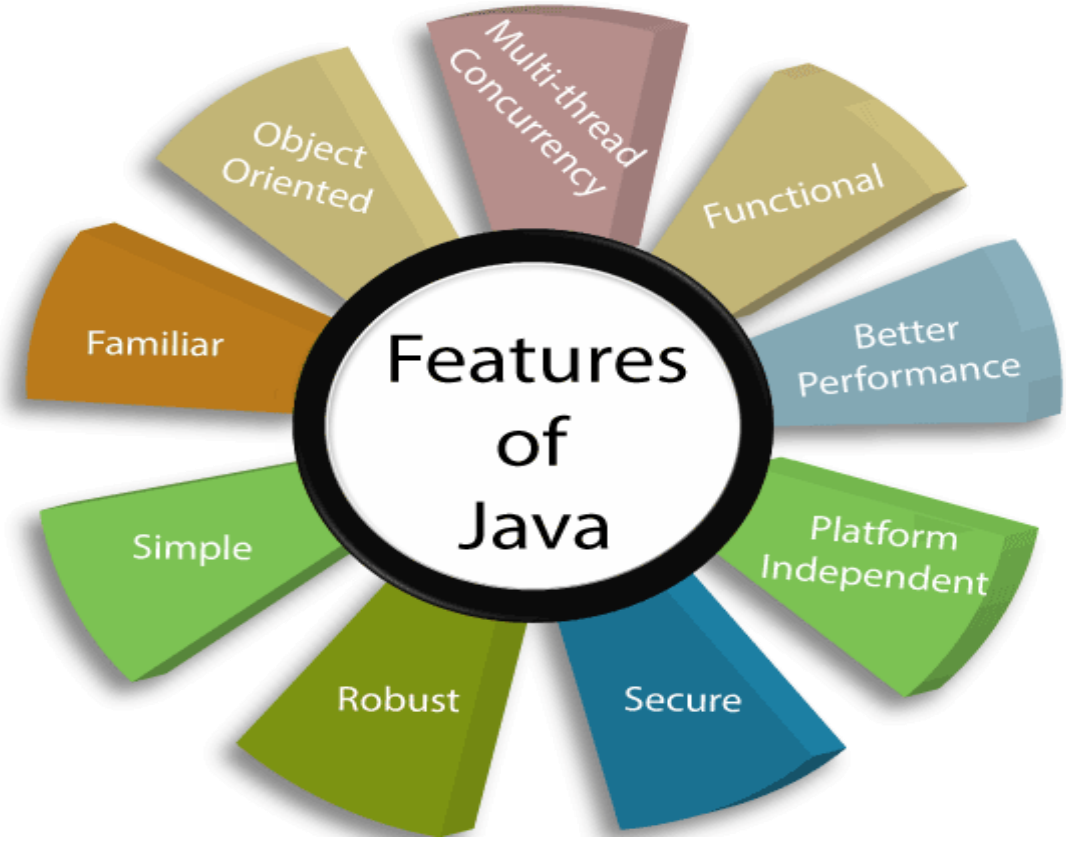
JAVA

Java is an object oriented programming language developed by **Sun Microsystems** in early **1990** by developers **James Gosling**, **Mike Sheridan** and **Patrick Naughton**. In 1991 James Gosling and his friends formed a team called **Green Team** to further work on this project. The original idea was to develop this programming language for digital devices such as television, set-top box, remote etc. Later this idea was dropped and Java is developed for internet programming

Original and first version of Java:

JDK first version alpha & beta was released in 1995 by Sun Microsystems.

Features of Java Programming Language



How is Java platform independent?

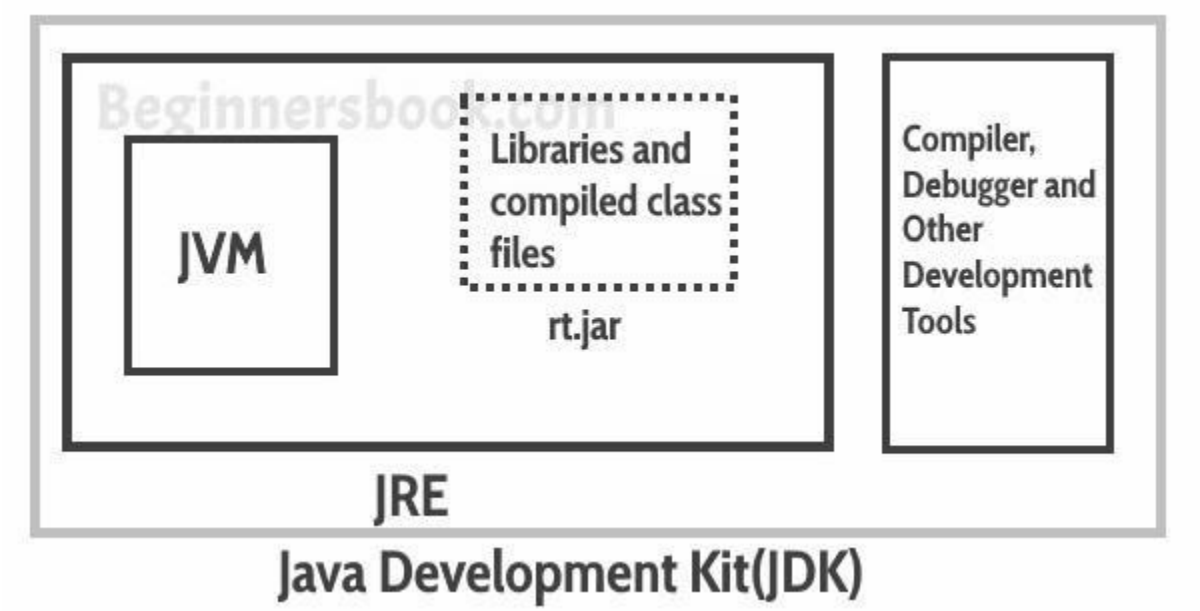
The meaning of platform-independent is that the java compiled code (byte code) can run on all operating systems. Three special things are introduced in Java: JDK, JVM, JRE…

JDK (Java Development Kit)

JDK is a superset of JRE, it contains everything that JRE has along with development tools such as compiler, debugger etc.

1. JDK stands for Java Development Kit. To install Java on your system, you need to first install JDK on your system.
2. JDK contains the tools that you need, in order to write and execute java programs.

3. JDK consists of Private JRE, JVM, compiler, Java application launcher, Applet viewer, etc. It also provides the Standard Edition (SE) of java API.



Features of JDK

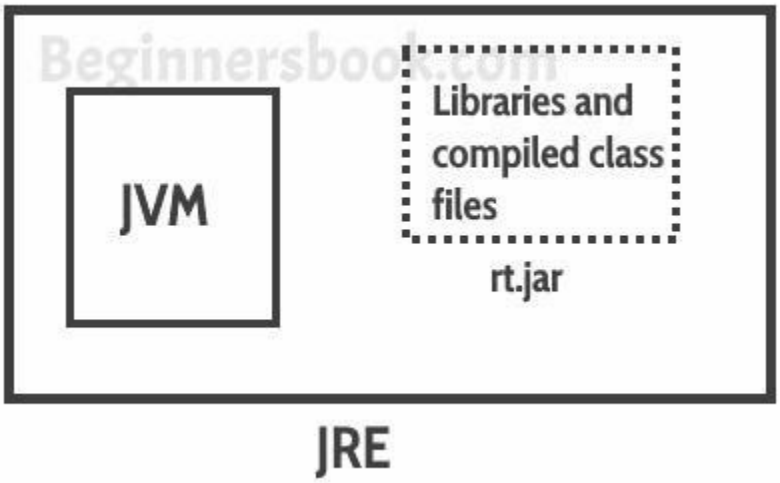
1. The JDK comes with a complete Java Runtime Environment (JRE), that is different from the regular JRE that is the reason it is usually called a private runtime so we can say that it includes all the features that JRE has.
2. It has all the java development tools such as compiler, JVM, JRE, debugger etc.
3. You need a JDK in order to write and run a java program.
4. JDK supports multiple platforms and can be installed on Windows, Mac and other operating systems.

JRE (Java Runtime Environment)

JRE is the environment within which the java virtual machine runs. JRE contains Java virtual Machine (JVM), class libraries, and other files excluding development tools such as compiler and debugger.
Which means you can run the code in JRE but you can’t develop and compile the code in JRE.

1. JRE stands for Java Runtime Environment. It provides a runtime environment for java applications.

2. You need JRE in order to run java programs. If you are not a developer and not writing java programs, you do not need JDK but you still need JRE to run java programs.
3. All the JDK already comes with JRE so you do not need to download and install it separately.
4. JRE contains a set of libraries and other files that JVM uses at runtime.



Features of JRE

1. JRE contains a set of libraries and other tools that JVM needs at runtime.
2. You can easily run any java program on JRE but you need JDK to write and compile java programs.
3. JRE contains libraries that are required for integrations such as Java Database Connectivity (JDBC), Java Naming and Directory Interface (JNDI), Remote Method Invocation (RMI) etc.

JVM (Java Virtual Machine)

Java is a high-level programming language. A program written in high level language cannot be run on any machine directly. First, it needs to be translated into that particular machine language. The **java compiler** does this thing, it takes java program (.java file containing source code) and translates it into machine code (referred as byte code or .class file).

Java Virtual Machine (JVM) is a virtual machine that resides in the real machine (your computer) and the **machine language for JVM is byte code**. This makes it easier for the compiler as it has to generate byte code for JVM rather than different machine code for

each type of machine. JVM executes the byte code generated by the compiler and produces output. **JVM is the one that makes the java platform independent**.

Simple Java Program

This is a very basic java program that prints a message “This is my first program in java”.

```
public class HelloJava {

	public static void main(String[] args)
	{
		System.out.println("hello Java");
	}

}
```

Output; hello Java

Public class HelloWorld

first line of our java program. Every java application must have at least one class definition that consists of `class` keyword followed by class name. When I say keyword, it means that it should not be changed, we should use it as it is. However, the class name can be anything.

I have made the class public by using public access modifier, I will cover access modifier in a separate post, all you need to know now that a java file can have any number of classes but it can have only one public class and the file name should be same as public class name.

Public static void main (String args [])

`public`: This makes the main method public that means that we can call the method from outside the class.

`static`: We do not need to create objects for static methods to run. They can run by themselves. `void`: It does not return anything.

`main`: It is the method name. This is the entry point method from which the JVM can run your program.

(String [] args): Used for command line arguments that are passed as strings. We will cover that in a separate post.

HOW TO USE SCANNER CLASS

program

```
public class Scanner Example {

        public static void main(String[] args)
        {

        Scanner in=new Scanner(System.in);
        System.out.println("enter a name");
        String name=in.next();
  System.out.println("name is :"+name);
        }

}
```

Output;

Enter a name
Java
Name is java

Input Types

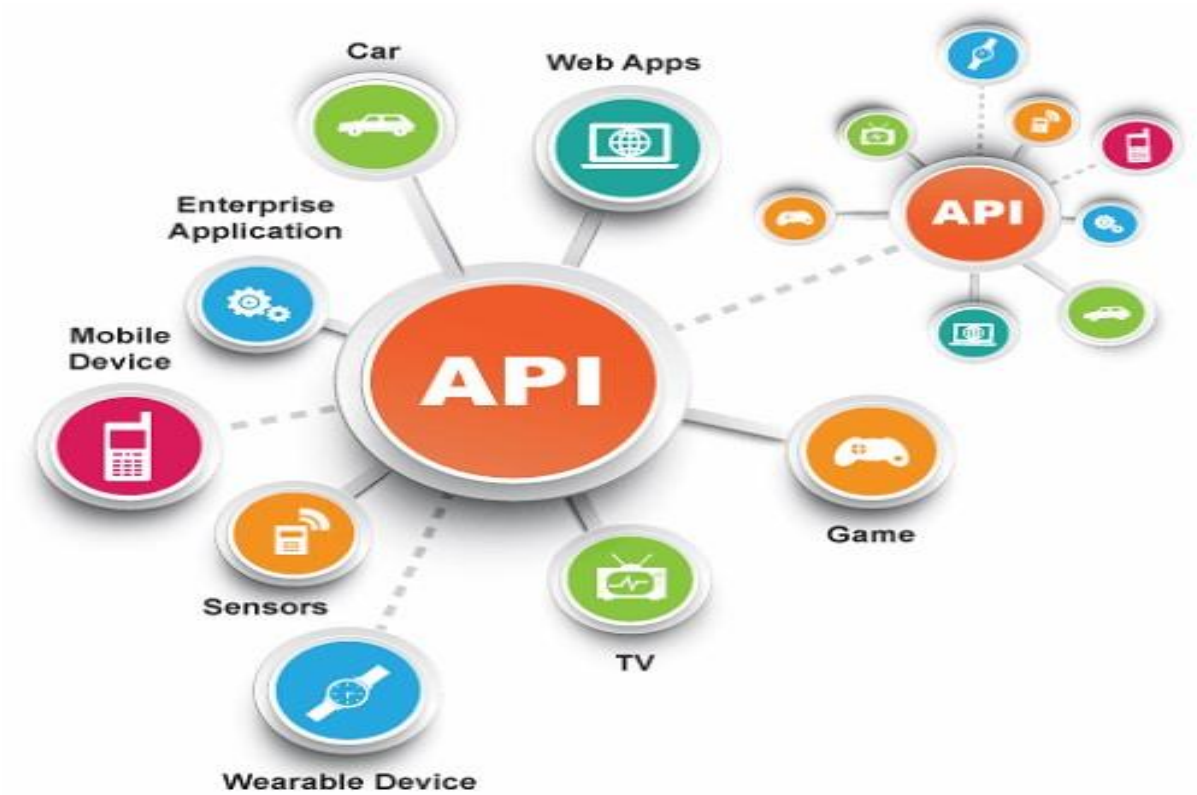
In the example above, we used the `nextInt (), next (), nextByte (), nextFloat (), nextDouble ()` method, which is used to read Strings. To read other types, look at the table below:

Java application

- Technology is constantly going through an evolution and so are the languages that are used to develop them. Java is one of the popular programming language having n number of applications. Through this blog, I will be listing down the top 10 applications of Java.

Desktop GUI Applications

- Mobile Applications
- Enterprise Applications
- Scientific Applications
- Web-based Applications
- Embedded Systems
- Big Data Technologies
- Distributed Applications
- Cloud-based Applications
- Web servers and Application servers
- Software Tools
- Gaming Applications



JAVA IS An OBJECT ORIENTED PROGRAMMING LANGUAGE (OOPS CONCEPT)

Object means a real-world entity such as a pen, chair, table, computer, watch, etc. **ObjectOriented Programming** is a methodology or paradigm to design a program using classes and objects. It simplifies software development and maintenance by providing some concepts:

- Object
- Class

- Inheritance
- Polymorphism
- Abstraction
- Encapsulation

Object

Any entity that has state and behaviour is known as an object. For example, a chair, pen, table, keyboard, bike, etc. It can be physical or logical.

Class

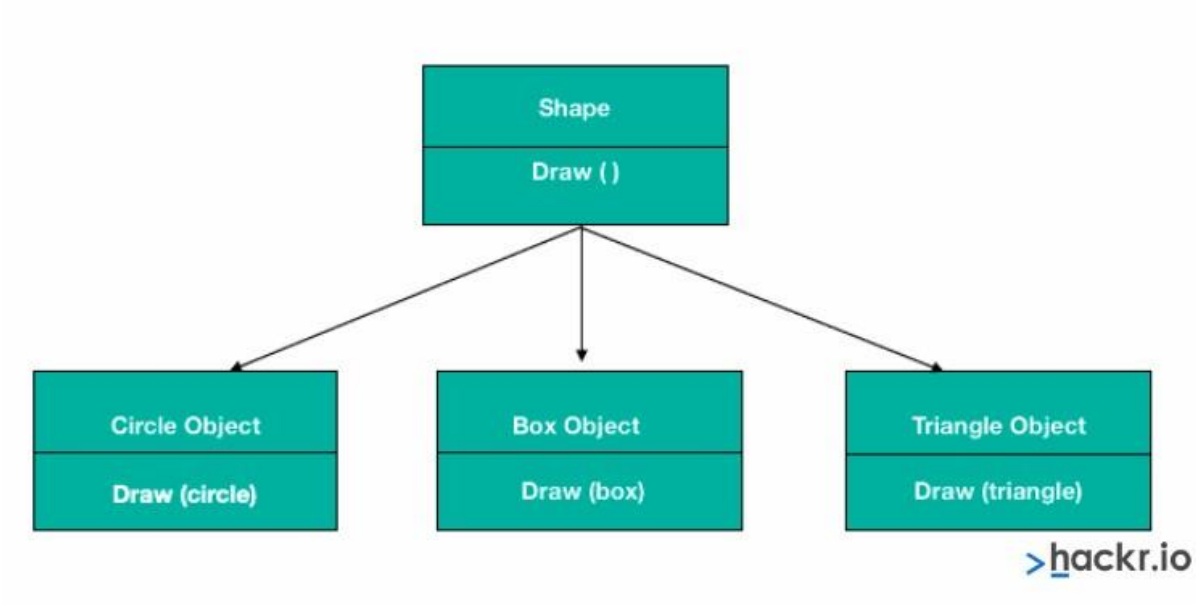
Collection of objects is called class. It is a logical entity.

A class can also be defined as a blueprint from which you can create an individual object. Class doesn't consume any space.

Polymorphism

Polymorphism is an important OOP concept; it means the ability to take many forms. For Example, an operation exhibits different behaviour in different situations. The behaviour depends on the type of data used in operation. For Example, in the operation of addition, the operation generates a sum for two numbers. If the operands are strings, then a third-string is produced by the operation by concatenation.

The figure below demonstrates that a single function name can be used to handle the different numbers and different types of arguments.



In polymorphism, objects having different internal structures can share the same external interface; it means that a class of operation may be accessed in the same manner even though actions with each operation may differ. Inheritance extensively uses the concept of polymorphism.

Polymorphism can be achieved in two ways: ○ **Method overloading** ○ **Method overriding**

Method declaration

```
package    methods;

public class Add1 {

      Add1()
      {
            System.out.println("java");
      }
      static int add(int a, int b)
      { int c=a+b;
            return c;
      }
      } package methods;

public class Constructor {

      public static void main(String[] args)
      {
            Add1 sum=new Add1();
            System.out.println(sum.add(20,10));
```

```
        }

}
```

Output;
java 30

Inheritance

When one object acquires all the properties and behaviours of a parent object, it is known as inheritance. It provides code reusability. It is used to achieve runtime polymorphism. They are five types

- Single inheritance
- Multiple inheritance
- Multilevel inheritance
- Hybrid inheritance
- Hierarchical inheritance

```
package shruthi;
class employee1 //parent class
{ double salary=700000;


        }

public class Inherit extends employee1 //child class
{ int bonus=10000; public static void
        main(String[] args)
        {
                Inherit emp=new Inherit();

                System.out.println("employee's salary is.. "+emp.salary);
                System.out.println("bonus is "+emp.bonus);


        }

}
```

Output;
employee's salary is... 700000.0
bonus is 10000

Abstraction

Hiding internal details and showing functionality is known as abstraction. For example phone calls, we don't know the internal processing.

Encapsulation

Binding (or wrapping) code and data together into a single unit are known as encapsulation. For example, a capsule, it is wrapped with different medicines.

A java class is an example of encapsulation. Java bean is the fully encapsulated class because all the data members are private here. **Example program**

```
package methods; public class

EncapsulationDemo {

        private int empId;
        private String empName;
        private int empAge;

public int getEmpId(
{ return empId;
}
public String getEmpName()
{ return empName;
}
public int getEmpAge()
{ return empAge;
}
public void setEmpName(String newValue)
 { empName=newValue;
}
public void setEmpAge(int newValue){
empAge = newValue;
}
public void setEmpId(int newValue){
empId = newValue;
}
}

package methods; public
class Encapsulation {

        public static void main(String[] args) {
                EncapsulationDemo caps=new EncapsulationDemo();
                caps.setEmpName("shruthi"); caps.setEmpId(4411);
                caps.setEmpAge(20);
                System.out.println("employee name is..."+caps. getEmpName ());
                System.out.println("employee age is..."+caps. getEmpAge
        ()); System.out.println("employ id is..."+caps. getEmpId ()); }

}
```

Output;

```
employ name is...shruthi
employ age is...20
employ id is...4411
```

Conclusion

Java offers the real possibility that most programs can be written in a type-safe language. However, for Java to be broadly useful, it needs to have more expressive power than it does at present.

This paper Addresses one of the areas where more power is needed.

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Certificate of Internship Completion

This is to certify that Kasnan .K from HINDUSTHAN INSTITUTE OF TECHNOLOGY, pursuing IT has successfully completed the internship in the field of JAVA from 28/12/2024 to 18/01/2025 in our organization. Throughout the **INTERNSHIP** program with us, He/She had been exposed to different processes and was found punctual, hardworking and inquisitive.

Trainer

Program co-ordinator

STUDENT INTERNSHIP DAILY REPORT AND FEEDBACK

Name of student	**:**	**KARNAN K**
Register Number	**:**	720823106039
Batch/Class	**:**	2023-2027 Batch-2nd year IT
Name of the Industry/Company	**:**	EMGLITZ TECHNOLOGIES PVT LTD, SAIBABA COLONY,COIMBATORE
Department(in which The training has been done)	**:**	Java Programming
Name of the Mentor (company)	**:**	LOGESHWARI
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